HumidClimatologyEngine v7.5

Bivariate Distribution Dominance Report Generator

Scientific documentation for the visual model-selection reporting layer

Purpose. The report generator is a deterministic visualization layer that consumes the frozen bivariate model-selection result and shows where and when each candidate joint-distribution family dominates. It does not refit distributions, alter model codes, or impose a Gaussian assumption.

1. Position in the v7.5 pipeline

```
Hourly ERA5-Land
    -> 5-day centered extraction
    -> candidate marginal models
    -> dependence / copula candidates
    -> empirical 2-D PDF reference
    -> model fit + AIC/AICc/BIC + diagnostics
    -> best_model_code(doy, latitude, longitude)
    -> bivariate_dominance_report.py
    -> maps + DOY heatmap + monthly shares + PDF report
```

The reporting layer is intentionally downstream of statistical fitting. This separation protects scientific reproducibility: the same selection product can be rendered repeatedly without recomputing or changing the fit.

2. Input contract

Input	Required structure	Meaning
best_model_code	doy x latitude x longitude	integer code of the selected bivariate model
model_names	global JSON/string attribute	code -> human-readable model name
latitude	latitude coordinate	spatial y coordinate
longitude	longitude coordinate	spatial x coordinate
month / day	optional coordinates	preferred calendar labels for summaries

The model mapping is part of the scientific provenance. Example codes may include Empirical-2D, Gaussian-Copula, t-Copula, Beta-Copula, or Bimodal-Copula, but the generator remains agnostic and reads the actual mapping from the selection product.

3. Report products

• **Monthly dominance-share chart.** Fraction of selected grid-day cells assigned to each model family for each month.

• **DOY x model heatmap.** For each climatological DOY, fraction of spatial cells selecting each model family.

• **Monthly spatial maps.** For each month, the mode of selected models over the DOYs assigned to that month, plotted across latitude/longitude.

• **Machine-readable PNG figures.** Exported alongside the PDF for papers, presentations, and QA archives.

4. Scientific meaning of dominance

A dominance map is a model-selection diagnostic, not a physical-truth map. It answers which candidate family most often wins the declared selection rule for a given day/location. The empirical 2-D PDF remains the non-parametric reference surface; any parametric joint model is a candidate approximation.

Recommended interpretation hierarchy

```
Empirical 2-D PDF
    |
    +-- reference surface from paired hourly observations
    |
    +-- parametric candidates
         |
         +-- marginal family (Normal / Skew-Normal / Pearson III / Beta / Bimodal)
         +-- dependence family (Gaussian copula / t-copula / other candidates)
         +-- information criteria + goodness-of-fit + tail checks
         +-- best_model_code
              |
              +-- visual dominance report
```

No Gaussian or other family is introduced by the reporting layer itself. If a model wins, that fact originates in the frozen fitting and selection stage.

5. Reproducibility contract

The generator should preserve: selection-product filename, model mapping, calendar convention, software version, configuration hash, input SHA-256 where available, and report creation timestamp. The report must be deterministic for a fixed input selection file.

6. Example execution

```
python bivariate_dominance_report.py \
    moisture_bivariate_model_selection_1981_2020_v7_5.nc \
    --output-pdf bivariate_distribution_dominance_report.pdf \
    --output-dir bivariate_dominance_figures
```

The command reads the selection NetCDF once, produces the PDF, and writes the reusable PNG figures. It does not modify the NetCDF selection product.

7. Quality-control requirements

• Confirm that model codes are finite where model selection is expected.

• Confirm that every model code has a valid name mapping.

• Confirm that monthly shares sum to approximately one when valid cells exist.

• Confirm that the DOY heatmap contains the expected climatological slot count.

• Confirm that spatial coordinates are monotonic and correspond to the selected product.

• Visually inspect at least the monthly maps, monthly-share chart, and DOY heatmap before publication.

• Archive the PDF and PNGs with the selection NetCDF and its checksum.

8. Release-bundle placement

```
HumidClimatologyEngine/
  moisture_climatology_v7_5.py
  bivariate_dominance_report.py
  README_v7_5.md
  scientific_reference.pdf
  main_climatology.nc
  diagnostics.nc
  bivariate_selection.nc
  bivariate_dominance_report.pdf
  bivariate_dominance_figures/*.png
  validation_log.txt
  provenance_manifest.json
```

This report generator is therefore part of the documented v7.5 scientific workflow, not an optional decorative post-processing script.